



Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Smart India Hackathon 2026 • Problem Statement PS-26183

Indian Cyber Crime Coordination Centre (I4C) • Ministry of Home Affairs, Government of India

Platform: CyberTrace Automated Analytics Engine

Target Focus: PS-26183 Solution Suite

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Version: 2.0 Production Ready

Repository: github.com/Hidayatulla268/CyberTrace

Live Demo: hidayatulla268.github.io/CyberTrace/

Executive Summary:

Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters in cases involving **task-based scams, sextortion, ransomware, phishing, fake job rackets, and organized cyber-enabled financial crimes**. CyberTrace solves this challenge through an end-to-end automated platform providing real-time multi-hop blockchain tracing, heuristic AI risk scoring, Centralized Exchange (CEX) attribution, zero-day Fraud DNA™ campaign matching, and court-ready Section 91 CrPC legal dossiers.

THE CORE FORENSIC WORKFLOW (USP)



1. Smart India Hackathon PS-26183 Context

Background: When cyber fraud victims lose funds to investment scams, task-based frauds, sextortion, ransomware, or phishing, the funds are swiftly transferred into cryptocurrencies. Fraudsters employ automated peeling chains, sub-minute splits, and intermediary layering wallets before depositing funds into Centralized Exchanges (CEXs) like Binance, WazirX, CoinDCX, KuCoin, or laundering through mixers.

The Operational Challenge for Law Enforcement:

1. *Velocity:* Funds move across 3 to 5 hops within minutes of victim reporting.
2. *Exchange Obfuscation:* Identifying whether an address is an intermediary or a known exchange hot-wallet requires multi-sig cluster heuristics.
3. *Zero-Day Wallets:* New fraudulent wallets that have not yet been reported on the 1930 portal escape traditional blacklist detection.
4. *Evidentiary Friction:* Investigators need structured, hash-verified Section 91 CrPC legal notices to freeze accounts before cash-out.

2. Comprehensive Feature Suite (13 Key Capabilities)

■ Smart Wallet Scanner	Instant multi-chain address scanner aggregating token balances (USDT, ETH, BTC, TRX, INR Equiv.), activity timelines, and full blockchain transaction ledgers.
■ Stolen Money Tracker	Interactive tranche follower that selects a victim deposit transaction (e.g. ■50,000) and follows that exact money hop-by-hop across peeling splits (60%/40% -> 36%/24%) with CSV export.
■ ■ Fund-Flow Graph	Interactive SVG visualizer mapping Victim -> Suspect -> Intermediary Splitters -> Exchange Clusters with live animated particle flows and node inspection.
■ Exchange Finder	Automated attribution of destination Centralized Exchanges (Binance Hot 14, WazirX, KuCoin, OKX) with confidence metrics (e.g. 91% Confidence) and legal nodal officer records.
■ Hidden Wallet Detector	Discovers unlisted accomplice infrastructure connected to the suspect, including Gas Sponsor Relayers, Co-Spending Mules, and Peeling Splitters with 1-click tracing.
■ Heuristic Risk Score	Dynamic 0–100 AI threat gauge categorizing wallets into Low, Medium, High, or Critical risk with clear forensic evidence bullet points.
■ Fraud Pattern Detection	Automated recognition of peeling chains, sub-45s script sweeps, fan-out/fan-in layering, and smart contract mixer calls.
■ Fraud DNA™ Detection	Breakthrough behavioral vector matching engine that identifies zero-day unknown scam wallets based on campaign DNA (91% match with Campaign #CYB-2048).
■ ■ Live Wallet Monitoring	Real-time mempool transaction ingestion stream with live pulse indicators, pause/resume controls, and instant threat alert triggers.

■ Fraud Reporting Portal	Victim complaint intake portal that accepts TXID, wallet address, amount, and scam type, auto-generating an FIR reference ID and immediately initiating trace flows.
■ Auto Evidence Report	1-click generation of court-ready Section 91 CrPC forensic dossiers containing full suspect metadata, flow graphs, timestamps, and SHA-256 integrity stamps.
■ Public Safety Check	Consumer pre-transaction scam screener providing instant SAFE, CAUTION, or ■ DO NOT SEND verdicts to prevent fraud before users send funds.
■■ Fraud Network Map	Cross-case crime syndicate graph correlating multiple independent victim complaints into an interconnected multi-case laundering nexus.

3. Fraud DNA™ — Detecting Scam Networks from Unknown Wallets

The Core Innovation: Traditional cybercrime tools rely on static blacklists. If a fraudster creates a new, unreported wallet (Wallet Z), blacklists give it a clean rating. **Fraud DNA™** analyzes the mathematical and behavioral sequence of transactions to identify if an unknown wallet behaves like a known crime syndicate.

BEHAVIORAL LINEAGE RECONSTRUCTION:

KNOWN FRAUD DNA → Wallet A (Victim 1 Hub) & Wallet B (Victim 2 Hub) → Synthesized Fraud Pattern → ■
Unreported Candidate Wallet Z (91% Match)

Display Alert:

■ **Potential Unreported Fraud Wallet Detected**

- **Wallet:** 0xAB89C41d2E5F78a9B30C2d4E6F8a91F2
- **Fraud DNA Match:** 91% Confidence
- **Attributed Campaign:** Campaign #CYB-2048 ("Hydra-Peel" Telegram Scam)
- **Forensic Evidence:** Similar 80/20 tranche split, sub-45s script delay, Binance sweep gateway, connected to 2nd-degree syndicate child node.

8-Dimensional Behavioral Vector Profile:

Vector Dimension	Analyzed Trait & Pattern	Weight
1. Transfer Cadence	Automated script execution velocity (<45s per hop)	94%
2. Peeling Ratio	Exact 80/20 & 60/40 fund-splitting tranche distribution	92%
3. Hop Topology	3-to-4 tier intermediary fan-out depth before sweep	96%
4. Tranche Sizing	Average victim tranche profile matching ■20k-■1L	88%
5. Destination Cluster	Consolidation into Binance Hot Cluster 14 gateway	91%
6. Gas Sponsorship	Shared relayer dispatcher funding gas via FixedFloat	86%

4. Law Enforcement Legal Compliance (Section 91 CrPC)

CyberTrace is designed to bridge the gap between blockchain analytics and courtroom admissibility under Indian law:

- **Section 91 CrPC Subpoena Automation:** Automatically drafts emergency freeze notices with exchange deposit memos and TXIDs.
- **Chain of Custody & Hash Integrity:** All generated evidence reports include SHA-256 cryptographic verification stamps.
- **Exchange Nodal Liaison:** Pre-mapped direct law enforcement desk emails for Binance, CoinDCX, WazirX, KuCoin, and OKX.

5. Step-by-Step Judges Live Demo Walkthrough

To evaluate CyberTrace in a hackathon or live investigation setting, follow the 7-step guided demo:

Step 1	Enter Suspect Address Input victim-reported suspect address or select a preset pill (e.g. Case #1245).
Step 2	Trigger Analysis Click 'Analyze' to crawl multi-hop transactions across EVM, Tron, and Bitcoin ledgers.
Step 3	Inspect Risk Score Review the 87/100 High Risk AI gauge and detailed 5-point evidence checklist.
Step 4	Explore Fund-Flow Graph Visually follow funds flowing from Victim -> Suspect -> Intermediaries -> Exchange Cluster.
Step 5	Detect Destination Exchange Identify Centralized Exchange deposit clusters (e.g. Binance Hot 14 with 91% confidence).
Step 6	Fraud DNA™ Zero-Day Test Click preset '■ Unreported Wallet Z' to trigger 91% behavioral match with Campaign #CYB-2048.
Step 7	Export Legal Dossier Click 'Generate Dossier' to print the court-ready Section 91 CrPC PDF evidence document.

6. Project Links & Quick Access

GitHub Repository:	https://github.com/Hidayatulla268/CyberTrace
Live Website (GitHub Pages):	https://hidayatulla268.github.io/CyberTrace/
Contact & Inquiries:	hidayatullashaik268@gmail.com